

AVEVA Academic Competition 2026 – Part 1

Background & Industry Context

As a process engineer, you've been given a new project to enhance your plant's cumene process. Cumene (isopropyl benzene) is the primary intermediate for phenol/acetone manufacturing and is produced by alkylating benzene with propylene over solid-acid catalysts according to the following reaction: $\text{C}_6\text{H}_6 + \text{CH}_2\text{CHCH}_3 \rightarrow \text{C}_6\text{H}_5\text{CH}(\text{CH}_3)_2$.

Before making changes in the plant, you'd like to build a simulation model of the process to use for optimization and running what-if scenarios. Most equipment in the process is easily modeled using appropriate thermodynamic methods and first-principles (aka equation-based) simulation. However, the kinetics of the cumene reactor are not well understood. This has historically blocked development of the simulation model. Recently, you've been hearing more about Artificial Intelligence and think that a combination of Machine Learning (ML) with first-principles simulation could help overcome this challenge.

To obtain training data for the ML model, your team recently collected detailed data from the process. Your task is to **develop a machine learning model trained on this real-world data (Part 1)** and **integrate it into AVEVA Process Simulation (Part 2)** to achieve a high-fidelity model that simulates the plant behavior.

During this project, you'd also like to improve operating data visibility for engineers, operators, and executives at the plant. You'll **design an interactive dashboard in CONNECT visualization** that shows reactor behavior, Key Performance Indicators (KPIs), and operational tradeoffs. This Cloud-based dashboard should be intuitive enough for operators to make informed decisions and clear enough for executives to understand the business impact.

Process Description and Process Flow Diagrams (PFDs)

Process Flow Diagrams (PFDs) for the cumene process are shown in **Figure 1** and **Figure 2** below. Benzene is combined with propene in a 5:1 ratio of benzene to propene. In HX-1005, the mixed feed is heated to 350°C using a DOWTHERM™ heat transfer utility. The hot feed stream then enters an adiabatic packed bed reactor (R-1100) to produce the cumene product. The reactor effluent is cooled to 90°C in HX-1102 before entering drum V-1104 to vent off light gases, which are used as fuel gas in other areas of the plant. The remaining benzene/cumene mixture is fed to a 13-stage distillation column (T-1200) where a 99 mol%

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Figure 1 - Process Flow Diagram (PFD) for the reactor section of the cumene process

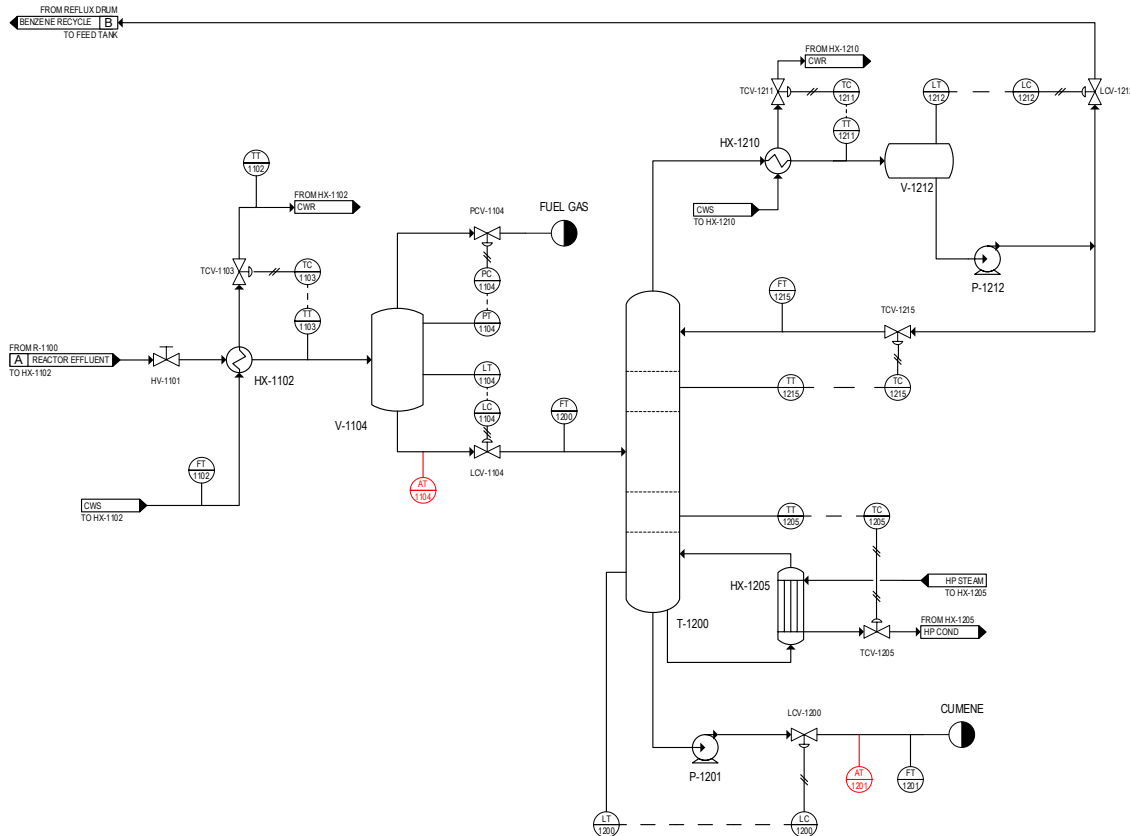


Figure 2 - Process Flow Diagram (PFD) for the separation section of the cumene process

Scope of Work – Part 1

In Part 1 of the AVEVA Academic Competition, you will focus on training a Machine Learning model to predict performance of the cumene process based on a set of inputs from the plant. You will also build an interactive dashboard that can be used by operators and executives at your company to monitor the process.

Data from your process is available in CONNECT data services. Your company has chosen CONNECT data services because it provides a secure, cloud-based platform for aggregating, storing, accessing, and analyzing real-time and historic data. You will use data measured between October 12, 2025 and October 17, 2025 to train your ML model. During this period, analytical transmitters were installed around the process to measure the density at various points, including at the outlet of the reactor. This will help you predict the conversion in the reactor and resulting composition of the reactor outlet stream. You were also given approval to ramp the process between a variety of operating conditions during this time.

Unfortunately, it is too costly to operate the analytical transmitters in normal production, and the process must now return to a normal operating state to meet existing production targets. Your task is to create a ML model that can predict the product quality of the reactor outlet stream so that the expensive physical transmitters can be avoided. **Table 1** shows data that is available to you in CONNECT data services. Using this information:

1. **Train an ML model** to predict the performance of the cumene reactor based on current process conditions.

NOTE: your Machine Learning model can be trained using the framework and algorithm of your choice. If you are just getting started, consider exploring scikit-learn in Python. There are numerous tutorials available online to help you get started. Whatever tool you choose, make sure your final model can be exported to ONNX format. This will be required when you use the model in AVEVA Process Simulation in Part 2 of the competition. For a full list of supported frameworks, visit onnx.ai/supported-tools.

2. **Create an interactive dashboard** in CONNECT visualization showcasing the available data from your process.

Deliverables – Part 1

Prepare a report (no more than 4 pages) describing the Machine Learning model you have trained and screenshots of your CONNECT visualization dashboards. The report should include:

1. A description of the model you chose for training (gradient boosting, nearest neighbors, etc.) and an explanation of why you chose the model.
2. A list of the input and output variables of your model and a description of how you chose those variables.

HINT: we have provided data from more transmitters than are required to train the model. Points will be awarded for selecting an appropriate subset of transmitters and justifying your decisions. When choosing your inputs and outputs, make sure to consider the problem you are trying to solve and the previous limitations that have prevented you from building a simulation model in the past.

Recall that all other equipment in the process (heat exchangers, distillation column, flash drums, etc.) can be accurately modeled using first-principles simulation. You

should consider the minimum set of information required to characterize the reactor outlet stream.

3. A summary of the metrics and scoring used to evaluate the accuracy of your model.
4. Screenshots of the dashboard(s) you have created in CONNECT visualization and a summary of how you expect those dashboards to be used in the plant.

NOTE: if you wish to prepare a video to accompany your screenshots, we will also accept a video (no more than 2 minutes) showing your dashboards and expected navigation experience.

Your report should be prepared using the following format:

- **Paper Size:** Letter (8.5 x 11in) or A4 (210 x 297mm)
- **Font Size:** 12
- **Font:** Aptos, Arial, Calibri, or Times New Roman
- **Line Spacing:** 1.15
- **Margins:** 1" (please include your last name in the header of the first page)
- **File Name and Format:** [LAST NAME]_2026AcademicCompetition_Part1.pdf

Grading Rubric and Submission Instructions – Part 1

Grading Rubric

Your submission will be graded according to the following rubric. Use the descriptions provided in the section above (Deliverables – Part 1) for more guidance.

Deliverable	Weight
ML model selection Points will be awarded for justification of the selected model, but not for the model selection itself. You may select any model if it is appropriately justified.	10%
Input and output variable selection Points will be awarded for justification of the selected input and output variables. While there is no single ‘correct’ answer, a data-driven approach should be taken that accounts for the provided data and constraints of the problem.	30%
Metrics and scoring Appropriate metrics should be provided that quantify the accuracy of your trained ML model. Be sure to also describe how you selected the training and test data for your model.	20%
CONNECT visualization dashboards Screenshots and/or video submission	30%
Formatting	10%

Submission Instructions

You will submit all deliverables for the competition after completing Part 2. You **do not** need to make a separate submission after completing the deliverables for Part 1. The problem statement for Part 2 of the competition will be released on **Friday, January 23, 2026**. Instructions will be provided for submitting your final deliverables.

Appendix

Table 1 - Measurement tag descriptions

Tag Name	Units	Description
AT-1100.PV	kg/m3	Density of reactor outlet stream*
AT-1104.PV	kg/m3	Density of column feed stream*
AT-1201.PV	kg/m3	Density of cumene product stream*
FT-1001.PV	kg/h	Mass flow rate of fresh benzene
FT-1002.PV	kg/h	Mass flow rate of benzene (includes recycle stream)
FT-1003.PV	kg/h	Mass flow rate of benzene (includes recycle stream)
FT-1005.PV	kg/h	Mass flow rate of DOWTHERM™ heat transfer fluid supply
FT-1102.PV	m3/h	Volumetric flow rate of cooling water
FT-1104.PV	m3/h	Volumetric flow rate of vent gases
FT-1200.PV	kg/h	Mass flow rate of column feed
FT-1201.PV	kg/h	Mass flow rate of cumene product
FT-1215.PV	kg/h	Mass flow rate of column recycle stream
FT-1300.PV	kg/h	Mass flow rate of benzene recycle stream
PT-1001.PV	kPa	Pressure of fresh benzene feed stream
PT-1004.PV	kPa	Pressure of propene feed stream
PT-1100.PV	kPa	Pressure of reactor outlet stream
PT-1300.PV	kPa	Pressure of benzene recycle stream
TT-1001.PV	°C	Temperature of fresh benzene feed stream
TT-1004.PV	°C	Temperature of propene feed stream
TT-1005.PV	°C	Temperature of DOWTHERM™ heat transfer fluid return
TT-1006.PV	°C	Temperature of reactor inlet stream
TT-1100.PV	°C	Temperature of reactor outlet stream
TT-1102.PV	°C	Temperature of cooling water return stream
TT-1103.PV	°C	Temperature after product cooler
TT-1211.PV	°C	Temperature of column overhead stream
FT-1004.PV	kg/h	Mass flow rate of propene feed stream
FC-1002.SP	kg/h	Benzene mass flow rate controller set point
FC-1002.PV	kg/h	Benzene mass flow rate
FRC-1004.SP	-	Mass flow ratio controller set point
FRC-1004.PV	-	Mass flow ratio (Benzene / Propene)
LC-1001.SP	%	Benzene feed tank level controller set point
LC-1001.PV	%	Benzene feed tank level
LC-1104.SP	%	Flash drum level controller set point

LC-1104.PV	%	Flash drum level
LC-1200.SP	%	Column sump level controller set point
LC-1200.PV	%	Column sump level
LC-1212.SP	%	Reflux drum level controller set point
LC-1212.PV	%	Reflux drum level
PC-1104.SP	kPa	Flash drum pressure controller set point
PC-1104.PV	kPa	Flash drum pressure
TC-1006.SP	°C	Reactor inlet temperature controller set point
TC-1006.PV	°C	Reactor inlet temperature
TC-1103.SP	°C	Product cooler utility outlet temperature control set point
TC-1103.PV	°C	Product cooler utility outlet temperature
TC-1205.SP	°C	Column bottoms temperature controller set point
TC-1205.PV	°C	Column bottoms temperature (measured at stage 11)
TC-1211.SP	°C	Column overhead temperature controller set point
TC-1211.PV	°C	Column overhead temperature
TC-1215.SP	°C	Column top temperature controller set point
TC-1215.PV	°C	Column bottoms temperature (measured at stage 3)

*Measurement is no longer available in the plant